Status	Finished
Started	Saturday, 17 October 2026, 5:58 PM
Completed	Saturday, 17 October 2026, 6:01 PM
Duration	2 mins 30 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. At the beginning of the list
c. In the right sublist
d. In the left sublist ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list
b. Compare the target element with the first element in the list
c. Compare the target element with the middle element in the list ✓
d. Divide the list into two equal parts

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search is more efficient than binary search on large lists.
c. Linear search requires the list to be sorted.
d. Linear search divides the list into halves.

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. log n
c. n-1
d. n ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By starting from the first element
c. By dividing the list length by two ✓
d. By comparing each element sequentially

Question **6**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search continues in the left sublist
c. The search ends successfully ✓
d. The search continues in the right sublist

Question **7**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It works on unsorted lists
c. It compares elements sequentially
d. It always starts from the beginning of the list

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search stops
- c. The search continues from the beginning
- d. The list is sorted

Question **9**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Deleting elements from a list
- b. Sorting elements in a list
- c. Determining whether an element is present in a list ✓
- d. Inserting elements into a list

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It can only be applied to large lists
- c. It is slower than linear search for small lists
- d. It requires the list to be sorted ✓

Question **11**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Hash
- b. Binary ✓
- c. Both (1) & (3)
- d. Linear

Question **12**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓

b. When the item is the last element in the array

c. When the item is not the array at all

d. Item is the last element in the array or item is not there at all

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O(n log n)

c. O(log n) ✓

d. O👎

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned

b. The first element is returned

c. An error is raised

d. The search is considered unsuccessful ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search

b. Linear search ✓

c. Binary search

d. Hashing